

Kyle D. Kavanagh

kdkavanagh@gmail.com | 248.914.8864 | Chicago, IL

EXPERIENCE

JUMP TRADING

HEAD OF MARKET STRUCTURE | JAN 2025 - PRESENT

- Responsible for forming and publicly advocating for Jump's opinions on market structure topics such as exchange design + implementation, product specifications, and regulatory proposals.
- Regularly interact with exchanges, regulators, and governments globally, conveying deeply nuanced technical topics to non-technical audiences.
- Lead internal initiatives to adapt Jump's trading systems to specific exchange nuance, in conjunction with trading teams, Performance Engineering, and Core Development.

LEAD QUANT DEVELOPER | TRADING TEAM | JAN 2025 - PRESENT

- Principle engineer for a mid-size futures trading team.
- Responsible for setting the direction for the team's research (python) and execution (C++) platforms and managing a team of experienced engineers.
- Authored the team's execution stack employing a custom python DSL/compiler atop a low-latency C++/CUDA engine, making it easy for researchers to test new features without sacrificing performance in production.
- Designed the team's Torch-based research platform with strong reproducibility guarantees and modular components that independently map onto Jump's HPC grid.
- Designed, implemented, and maintain the firm's most popular grid-compute tooling, allowing researchers to easily fan compute jobs out across a Slurm cluster. Adoption grew by word-of-mouth to 500+ users firmwide.

LEAD QUANT DEVELOPER | CENTRAL TRADING & RESEARCH | JULY 2019 - JAN 2026

- US lead for 'Central Trading & Research', a "special projects" and central trading team responsible for microstructure research and proprietary feed design.
- Designed Jump's exchange-feed normalizations used in MF/LF contexts, prioritizing information content and ease-of-use over lowest-latency.
- Pioneered a new data warehouse paradigm across the firm, allowing researchers to work with datasets more friendly than raw PCAPs.
- Led the design and implementation of Jump's USEQ ISO trading framework.

COINBASE | MATCH ENGINE PRODUCT AND DEVELOPMENT | SR TRADING SYSTEMS ENGINEER

April 2018 - April 2019 | Chicago, IL

- Lead distributed systems architect and engineer for Coinbase Markets' low-latency electronic trading platform.
- Designed a full-featured exchange architecture from first-principles. The architecture provided low-latency operation with full fault tolerance and 24x7 capabilities. Designed bespoke Order Entry and Market Data protocols and related messaging libraries as interfaces to the exchange. Authored an event-sourced Matching Engine for the new platform.
- Helped open the Chicago office and recruit and hire over twenty people across product, software, x86, networking, and physical infrastructure to build a fully self-sufficient exchange operation.

CME GROUP | MATCH ENGINE DEVELOPMENT | LEAD SOFTWARE ENGINEER

July 2014 - April 2018 | Chicago, IL

- Served as the Globex team's representative to the COO's cross-functional Market Microstructure Working Group, responsible for shaping the direction of CME's markets and defining CME's official positions on market structure topics.
- Responsible for improving the performance and capacity of CME's low-latency Globex Electronic Trading Platform.
- Designed and led the implementation of the architecture for CME's original (internal) hybrid cloud infrastructure.

EDUCATION

UNIVERSITY OF MICHIGAN

COLLEGE OF ENGINEERING

MAY 2014 | ANN ARBOR, MI

B.S.E Computer Science, Cum Laude
Financial Mathematics Minor

PATENTS & AWARDS

2015-2025

27 granted patents, 14 currently pending relating to electronic trading, software and network performance, messaging optimization, and UI/Design.

2008

Eagle Scout